

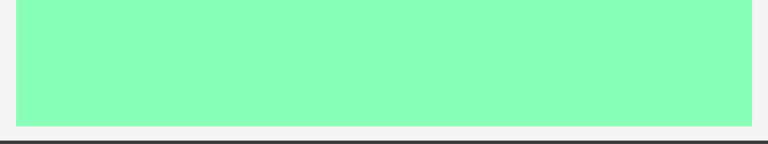
BIOLOGY 30

UNIT 2:
**Reproduction &
Development**



Chapter 15

Human Development



Chapter 15: Human Development

- Reproductive technologies
- Fertilization
- Cleavage, gastrulation & implantation
- Neurulation & organ development
- Supportive structures (placenta, amniotic cavity)
- Stages of fetal development
- Environmental factors & fetal development
- Parturition & lactation (role of prolactin & oxytocin)

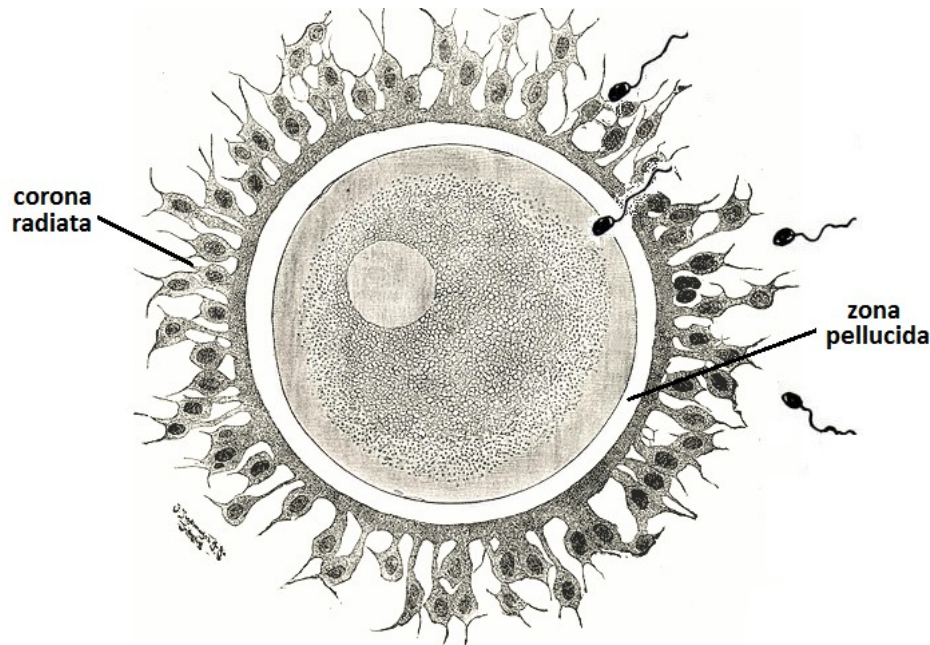
Textbook pgs. 506-543

Fertilization



Fertilization

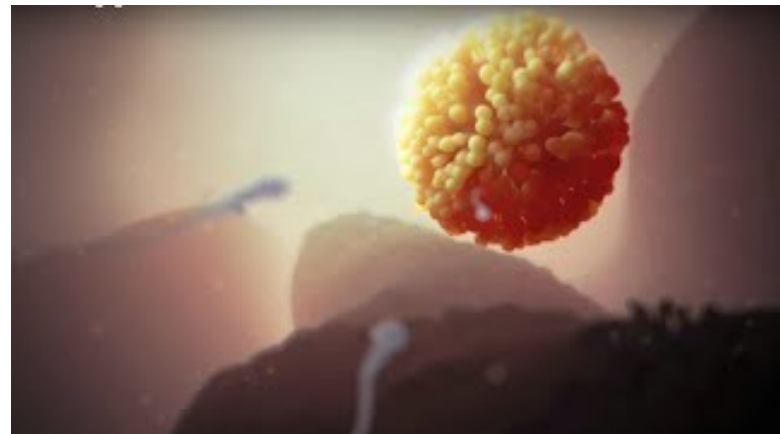
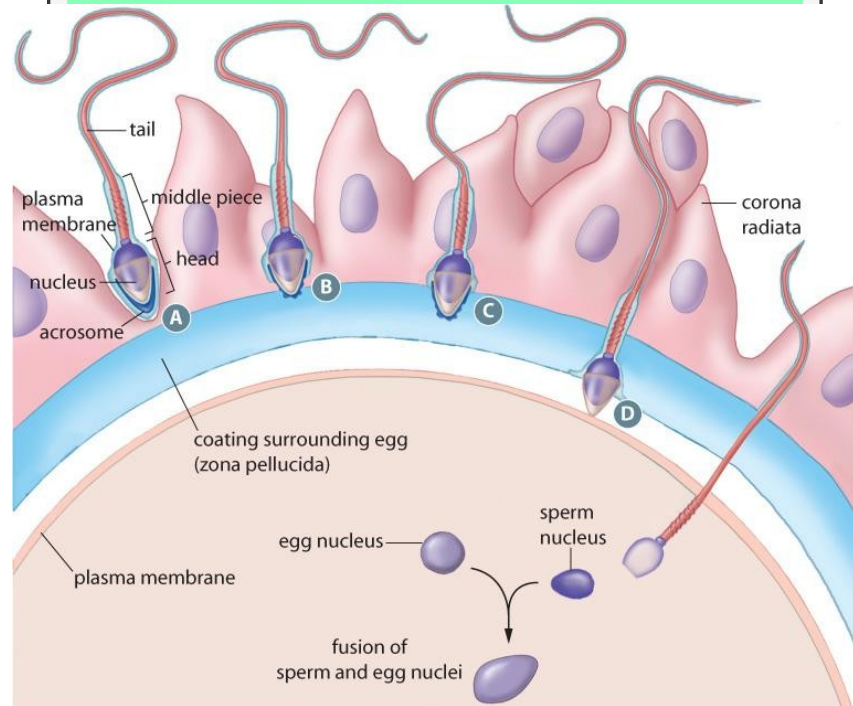
- Fertilization is the **joining of male and female gametes** (sperm and egg) to form a **zygote** (a fertilized egg).
- Each **gamete is haploid**, meaning they contain only 23 chromosomes or half the amount other cells have. As these two haploid cells fuse, they form a **diploid zygote** with 46 chromosomes. (More on ploidy and cell division in Unit C)



- **Occurs within the oviduct (fallopian tubes)**
- Unique coating layers of the egg (corona radiata and zona pellucida) ensure that only one sperm may successfully attach

THE PROCESS:

1. Sperm cells release enzymes from their acrosome cap to digest the outer coating of the egg
2. Sperm cells move farther through the coating towards the plasma membrane
3. Once a sperm enters the egg, the egg's plasma membrane depolarizes, preventing other sperm from entering
4. The nuclei of the sperm and egg fuse, resulting in a single cell with a complete set of 46 chromosomes (referred to as the *zygote*)



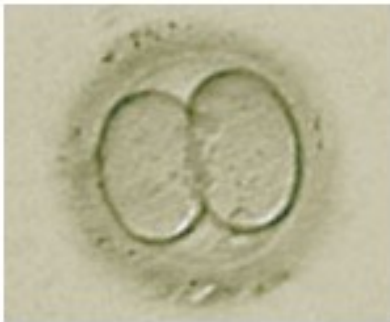
Stages of Development

- Pregnancy can be divided into:
 - 2 stages
 - Embryonic stage: weeks 1-8
 - Fetal stage: weeks 9-12
 - OR**
 - 3 trimesters
 - First trimester: months 1-3
 - Second trimester: months 4-6
 - Third trimester: months 7-9

Important developmental milestones occur during these periods.

Cleavage and Implantation

After fertilization, the zygote continues moving through the oviduct towards the uterus. During this time, several cell divisions occur, leading to the formation of the morula.



2-cell zygote



4-cell zygote



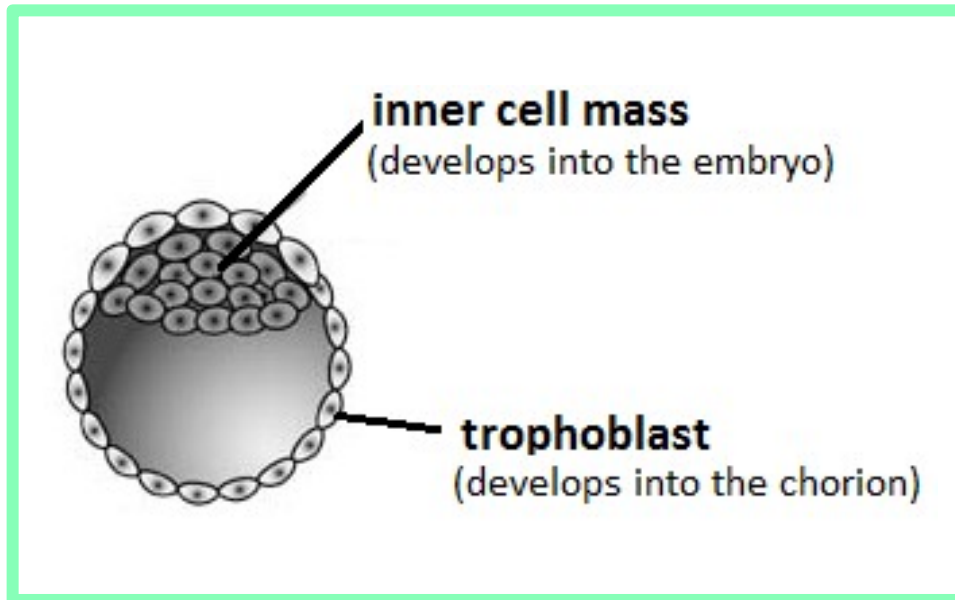
8-cell zygote



morula

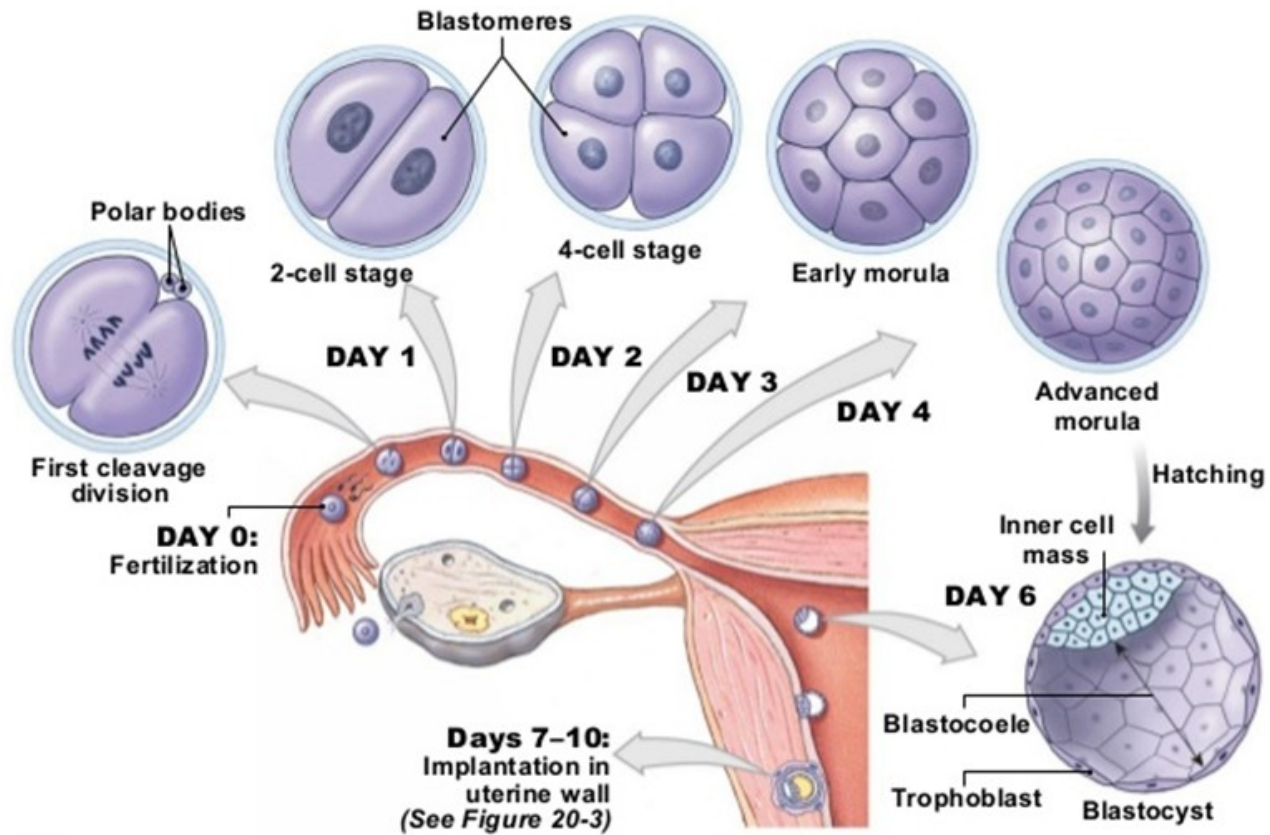
- Initial cell division occurs rapidly, leaving little time for cell growth. This process of cell division without enlargement is called **cleavage**.
- Over the 3-5 day period required to reach the uterus, cleavage occurs several times to form a sphere of 16 cells. This is the **morula**.

- Upon entering the uterus, the *morula* further develops into the **blastocyst**, a fluid-filled structure consisting of two layers:
 - the **trophoblast** which will release hCG and later turn into the chorion and placenta, and
 - and the **inner cell mass**, which will later develop into the embryo

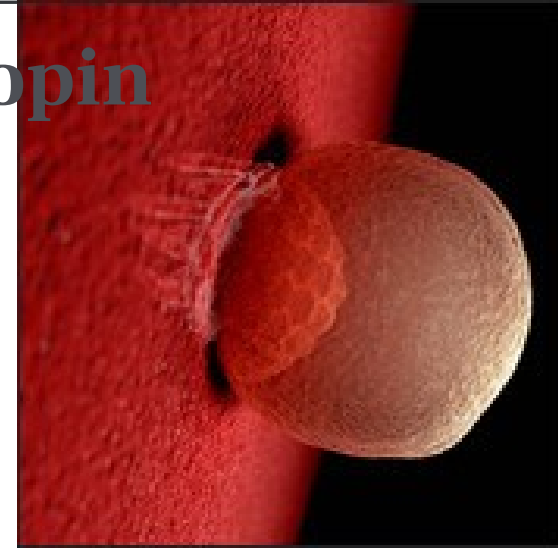


- Between days 5-7, the blastocyst will begin attaching itself to the endometrium of the uterus; a process known as **implantation**.
- Implantation is complete by days 10-14. If successful, the woman is now said to be pregnant.

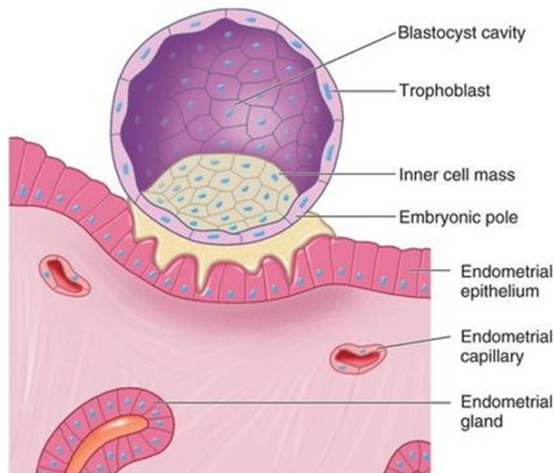
Figure 20-2 Cleavage and Blastocyst Formation.



Human Chorionic Gonadotropin (hCG)



- Remember... in the menstrual cycle, the role of the corpus luteum is to release progesterone (and some estrogen) to maintain the endometrial lining should implantation occur. If no fertilization and implantation occur, the corpus luteum degenerates after about 14 days. In turn, there is no progesterone and estrogen produced and thus the endometrium will shed → menstruation.



(a)

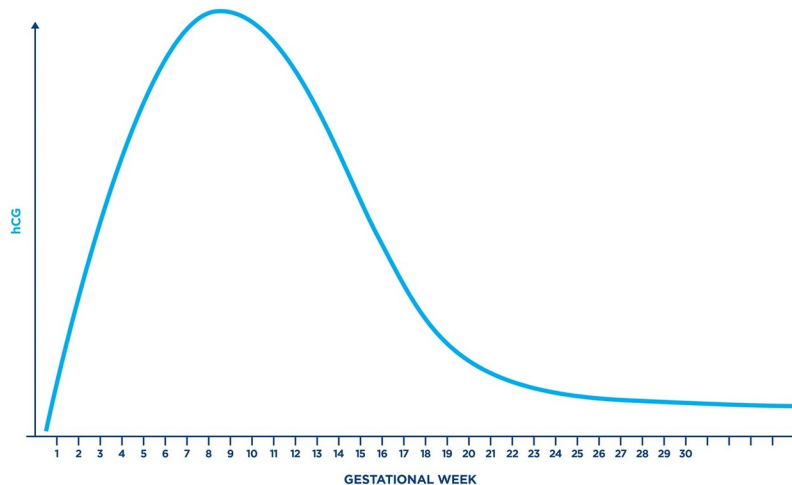
But what if fertilization and implantation do occur?

We do not want the endometrium lining to shed as that would lead to a miscarriage.

Human Chorionic Gonadotropin (hCG)

- Once implantation occurs, the trophoblast begins secreting the hormone human chorionic gonadotropin (hCG)...

hCG LEVEL DURING PREGNANCY



- hCG maintains the corpus luteum so that it continues secreting estrogen and progesterone
- This maintains the endometrium and prevents menstruation
- hCG levels decline after the second month of pregnancy, when the placenta is able to secrete these hormones on its own

hCG Pregnancy Tests

- Urine pregnancy tests look for the presence of hCG. Since the trophoblast is the only possible source of hCG in the body, if hCG is present then pregnancy is confirmed.
- Since the trophoblast begins to produce hCG within 1-2 weeks of fertilization, which occurs soon after ovulation, hCG can be often detected up to one week before an expected menstrual cycle. Most accurate results, however, would be expected after a missed cycle.



How to read results

Negative result

One pink line



Positive result

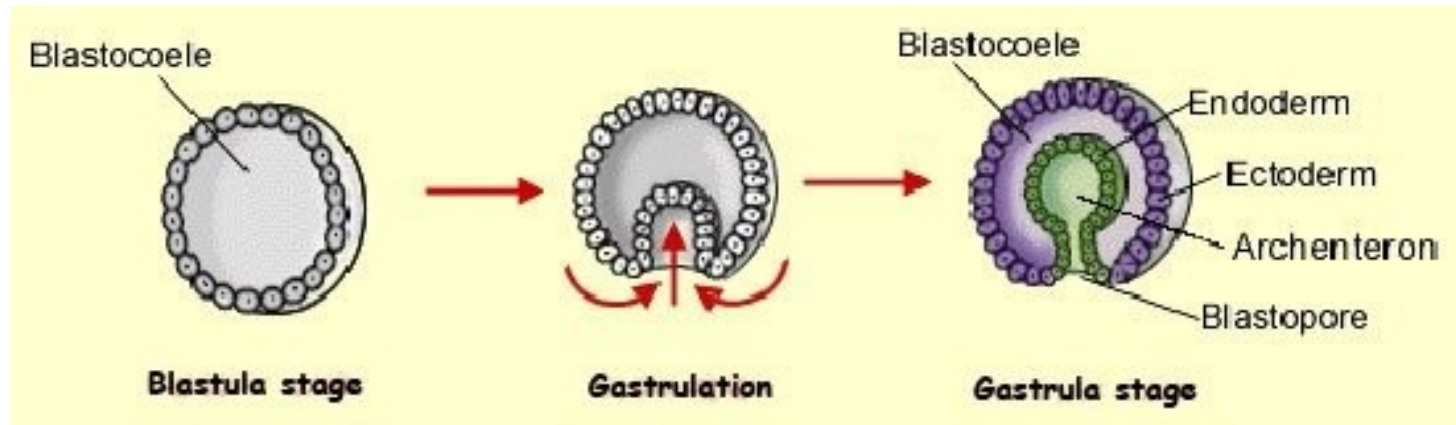
Two pink lines



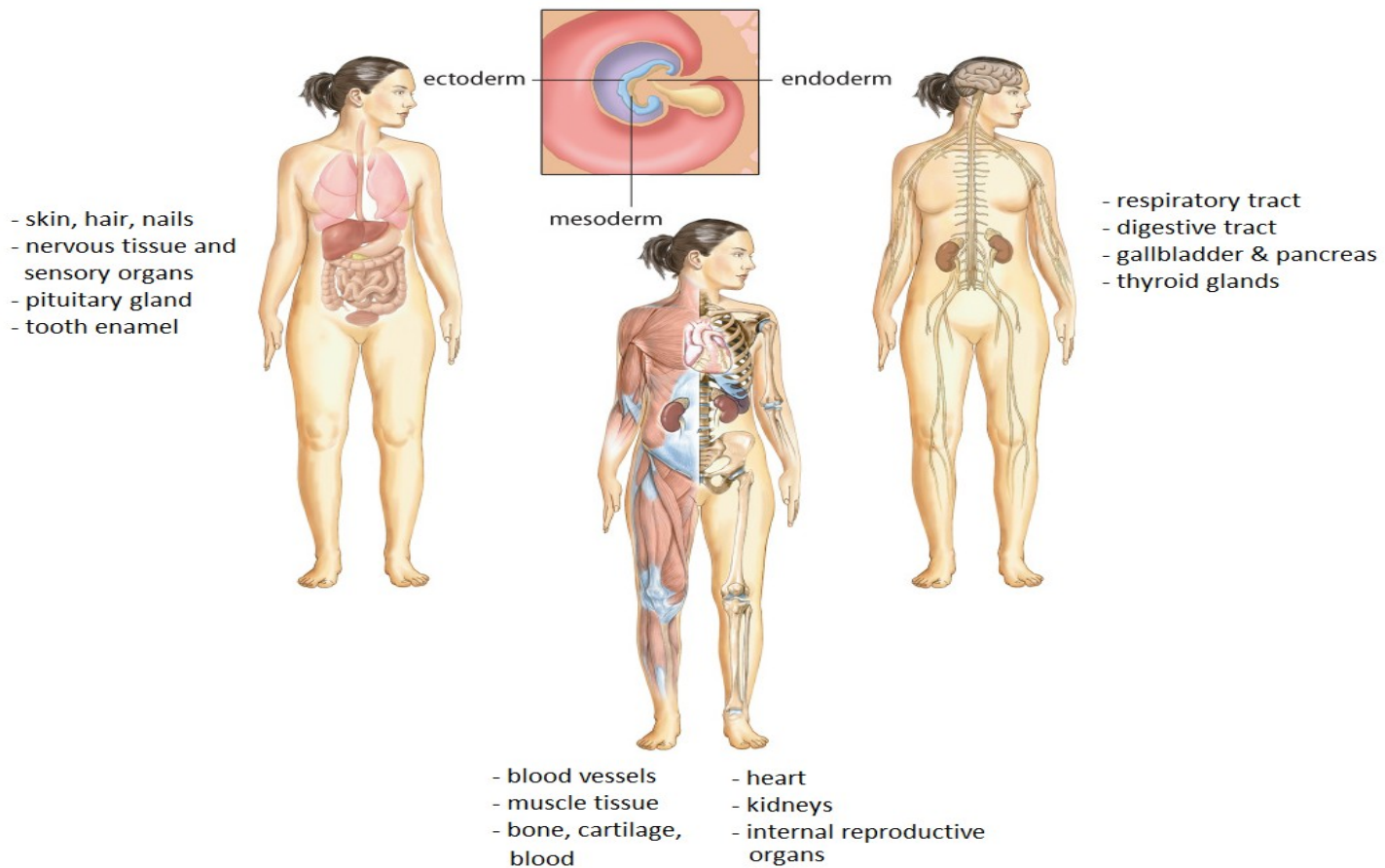
Gastrulation

During week 2, the blastocyst begins to change...

- A space known as the ***amniotic cavity*** forms between the inner cell mass and the trophoblast.
 - This cavity will contain **amniotic fluid** that will serve to cushion and protect the developing baby from physical trauma
- The inner cell mass flattens and folds upon itself to create three distinct layers: the **outer *ectoderm*, the inner *endoderm*, and the middle *mesoderm***.
- This process is known as ***gastrulation***, after which the blastocyst is referred to as the ***gastrula***.



Each of the three layers (referred to as germ layers) differentiates into a specific set of tissues and organs:





GASTRULATION

The second phase of development is gastrulation, a migration of blastocyst cells inward, leading to multiple distinct layers of tissue called germ layers.

GERM LAYER

Ectoderm

Endoderm

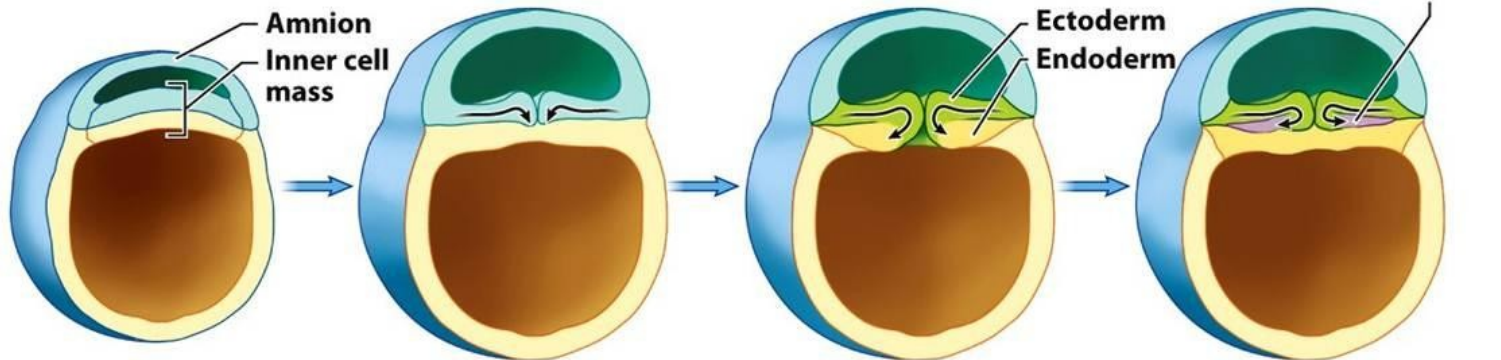
Mesoderm

BECOMES

Outer layer of skin, hair, lining of the nose and mouth, and the nervous system

Digestive tract, respiratory tract, liver, and pancreas

Muscles and skeleton



Cells of the inner cell mass begin to differentiate as the amnion forms

A gastrula develops when cells begin to migrate inward, forming an indentation.

The cells continue to push inward, forming the endoderm. Cells that remain on the outer surface of the gastrula are called ectoderm.

The mesoderm is formed, as additional cells migrate inward between the endoderm and ectoderm.

Structures that support the embryo

- Starting in week 3 up to week 8, membranes external to embryo develop, called **extraembryonic structures**

Allantois

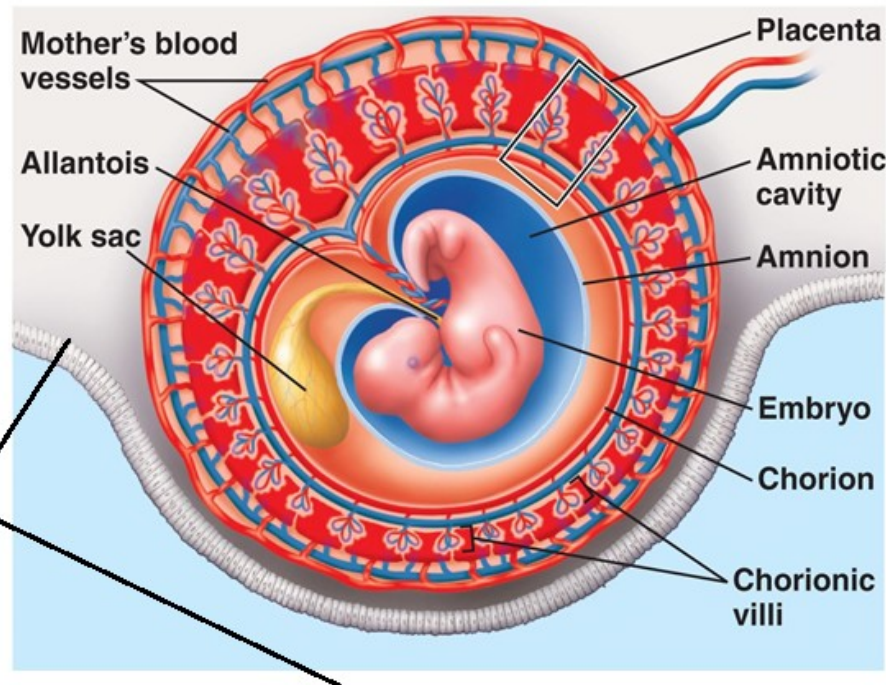
Forms the foundation for the umbilical cord. Degenerates during the second month

Yolk sac

Forms the digestive tract. Shrinks after week 8.

Also produces the first red blood cells until the bone marrow is formed and takes over.

These extra-embryonic membranes are expelled from the uterus as afterbirth



Placenta

Attaches the fetus to the uterine wall. Where metabolic exchange occurs. Composed of embryonic and maternal blood vessels.

Amnion

Encloses the fluid-filled sac which protects the embryo during development

Chorion

Outermost layer. Forms the fetal portion of the placenta.

Organ Formation (Organogenesis)

Occurs between weeks 3 & 8, when germ layers begin to fold and early organ systems take shape...

Week 3 = neurulation (formation of **neural tube** which will later develop into the brain and spinal cord), primitive heart starts beating

Week 4 = Formation of blood cells. Lungs and kidneys take shape, and small buds (which will develop into arms and legs) appear. A distinct head is visible.

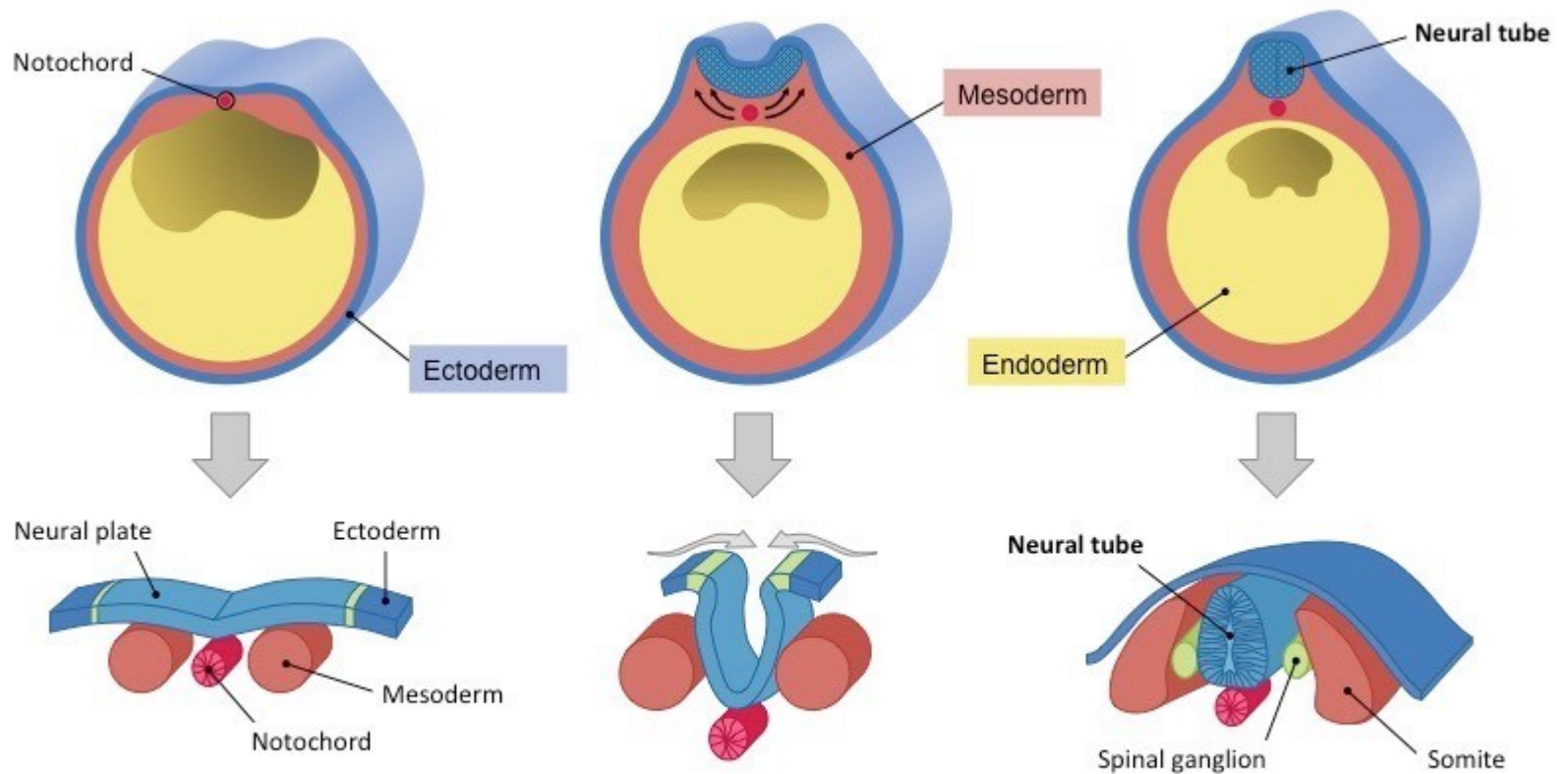
Week 5 = Eyes open, cells of the brain undergo rapid differentiation.

Week 6 = Continued development of the brain, lengthening of limbs.

Week 7 = Organs and skeleton almost completely formed, external genitalia begin to differentiate. Nervous system begins to coordinate body activity

Week 8 = Embryo is the size of a paperclip. 90% of organs are established.

Neurulation



1. Notochord forms from mesoderm cells soon after gastrulation is complete

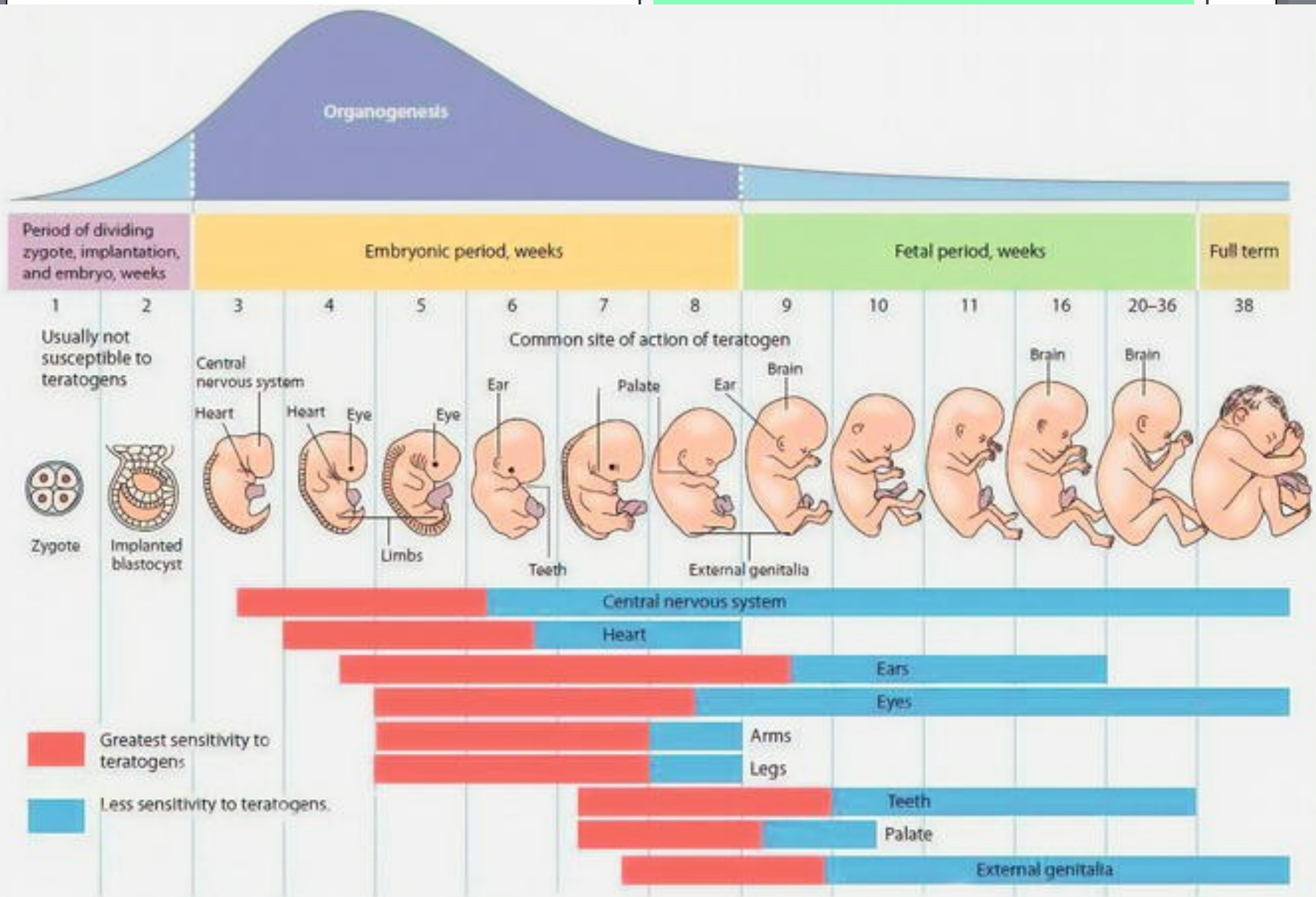
2. Signals from notochord cause inward folding of ectoderm at the neural plate

3. Ends of neural plate fuse and disconnect to form an autonomous neural tube



FUNCTIONS OF THE PLACENTA:

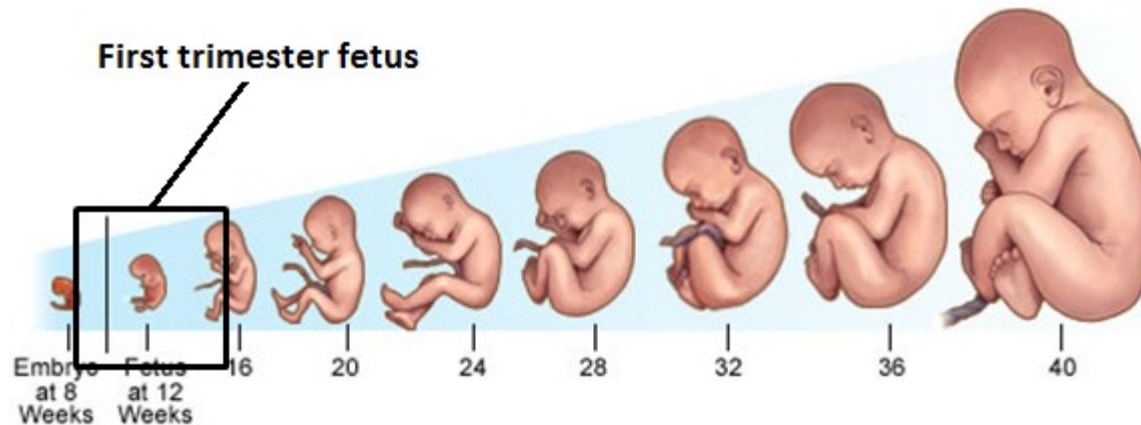
Nutritional functions	• Transports nutrients from mother to fetus • Stores nutrients
Excretory functions	• Transports wastes from fetal blood to the mothers blood
Respiratory functions	• Transports oxygen from the mother to the fetus and CO₂ from the fetus to the mother
Endocrine functions	• Secretes hormones • Allows hormones from the fetus to diffuse into the mothers blood (and vice versa)
Immune functions	• Transports antibodies from the mother to the fetus



Fetal Development

FIRST TRIMESTER

(weeks 1-12)

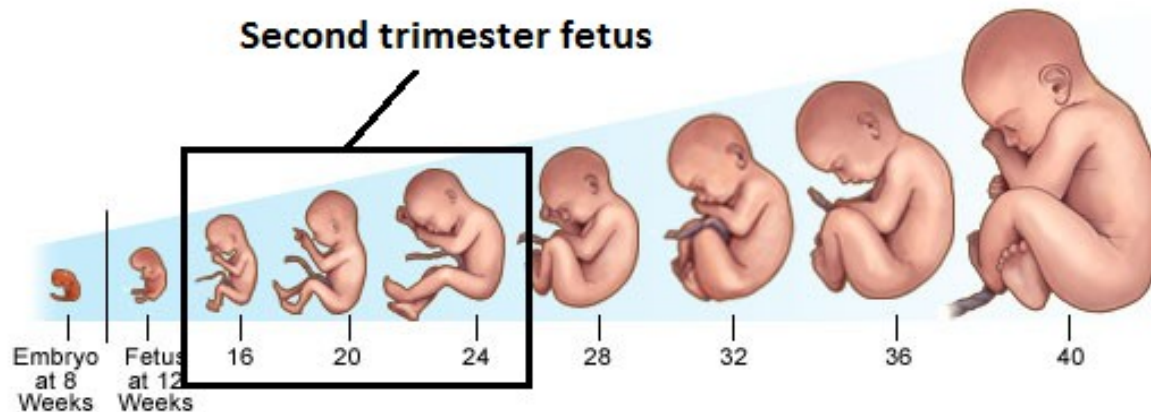


- embryonic period = weeks 1-8
 - gastrulation and organ differentiation)
- fetal period = weeks 9-12
 - Body continues to grow, skeleton begins to harden
 - External reproductive organs become distinguishable

Fetal Development

SECOND TRIMESTER

(Weeks 13-24)

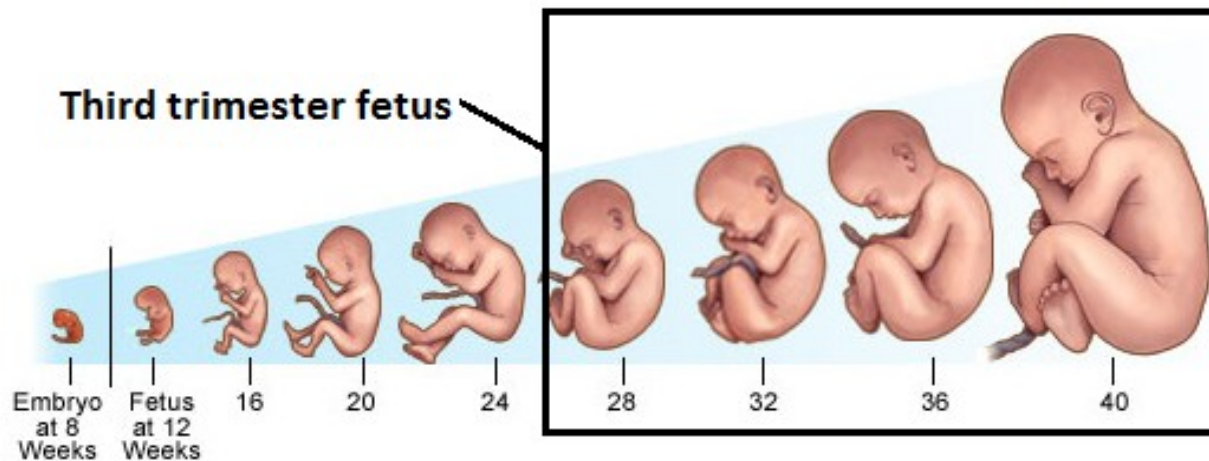


- Heartbeat is strong enough to hear with a stethoscope
- Rapid brain growth, nervous system begins to function
- Mother begins to feel movement
- Fetus would not likely survive if it was born at this stage

Fetal Development

THIRD TRIMESTER

(weeks 25-38)



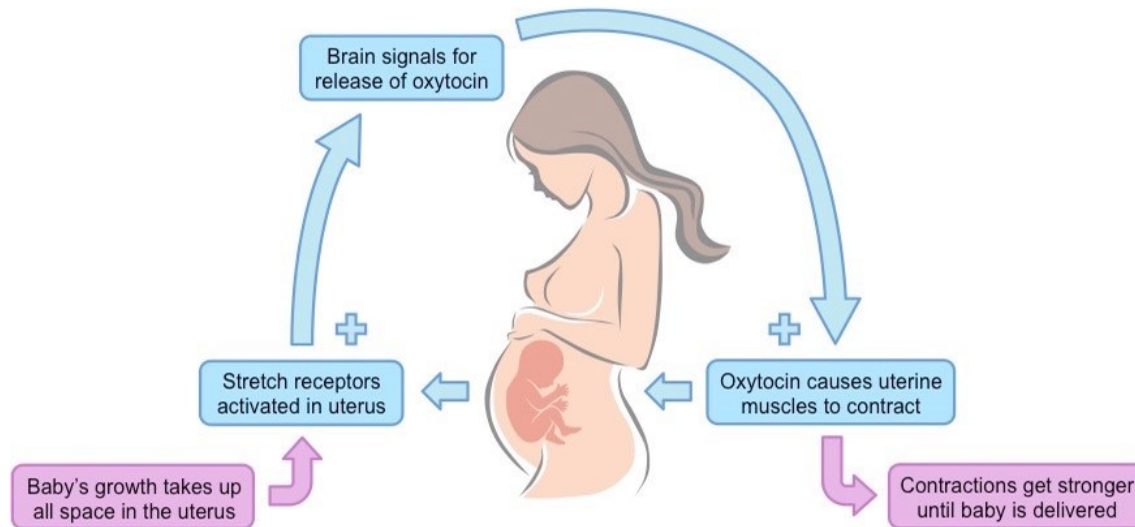
- Rapid brain development and organ maturation
- Fetus nearly doubles in size
- Month 7: fetus turns upside down and usually survives if born at this stage

From Conception to Birth



Parturition: Delivery of the Baby

The onset of labour, or uterine contractions, is regulated by **positive feedback mechanisms...**



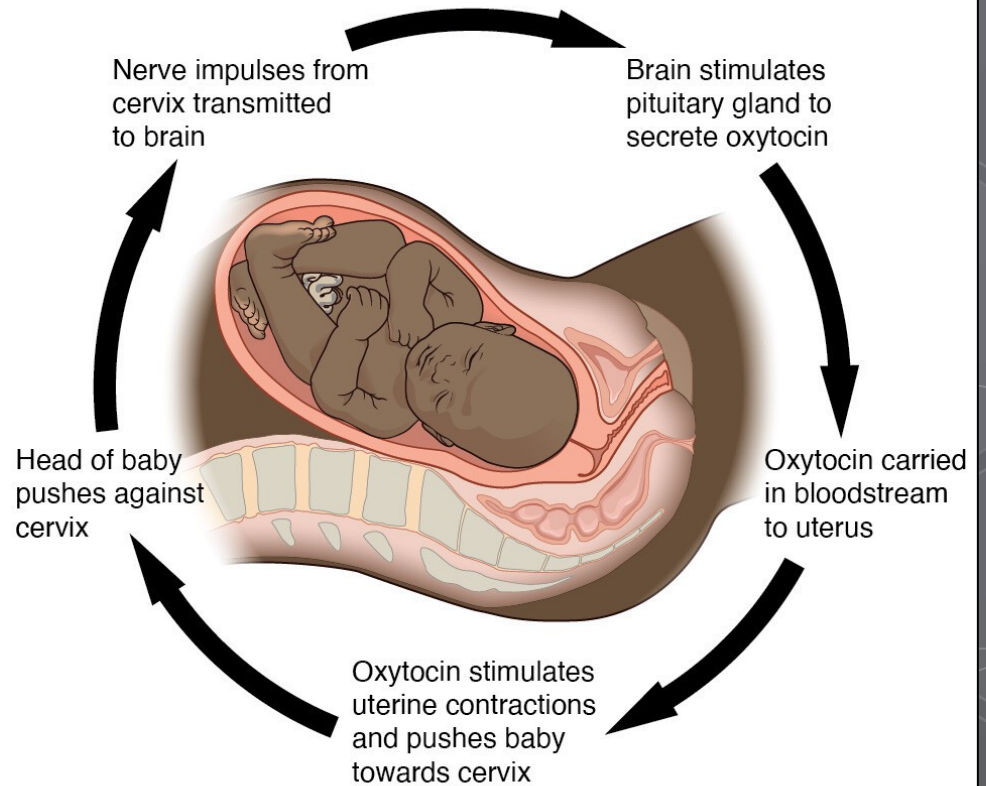
- The cycle continues until birth occurs
- Contractions are also aided by the production of **prostaglandins**, a group of hormones which act on the same tissue they are produced by

Parturition



Repeating cycle - no return to homeostasis

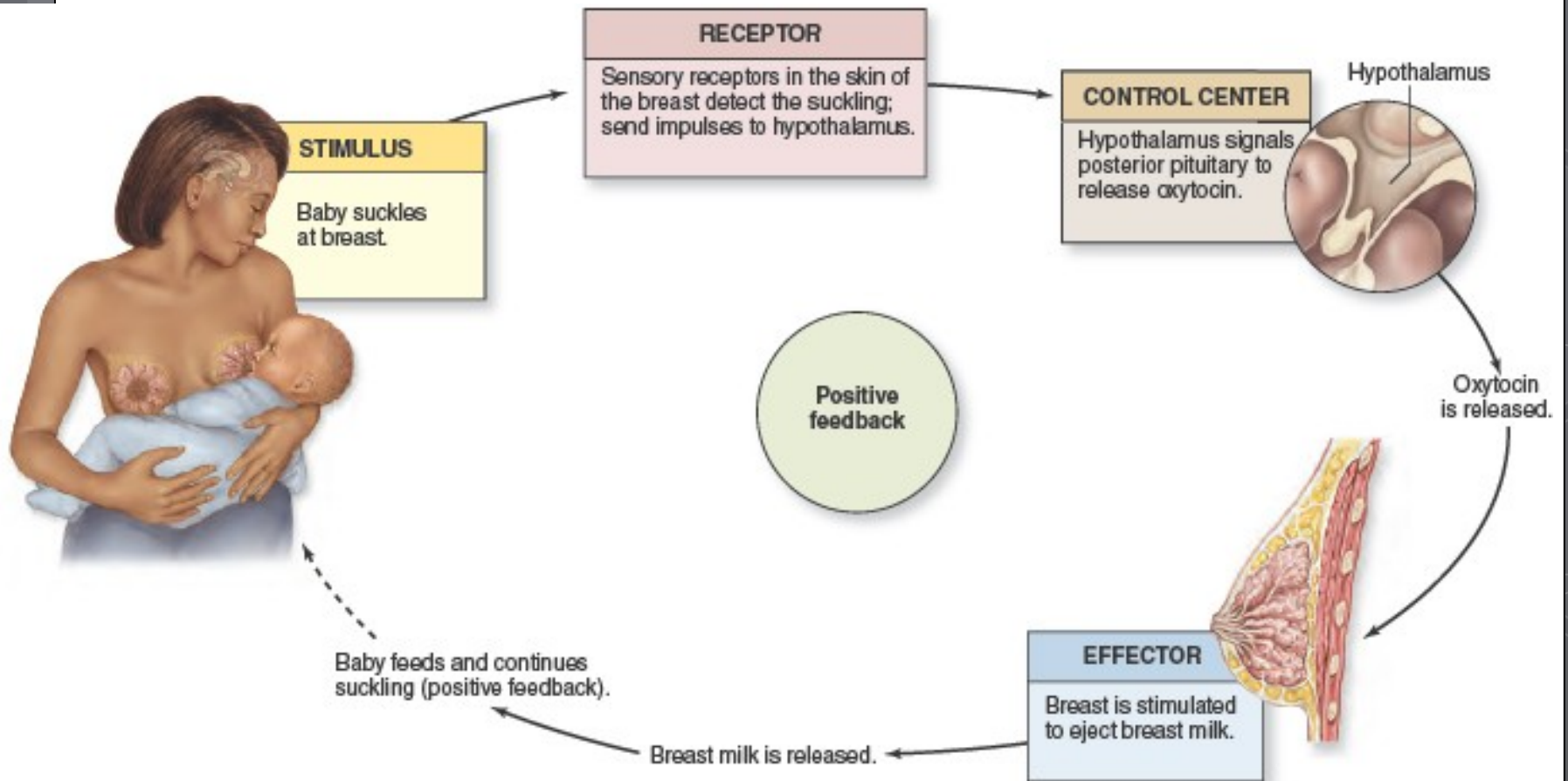
- The feedback cycle repeats with greater speed and intensity until the cervix is fully dilated and the baby is born.



Hormonal control of lactation:

- During pregnancy, high levels of estrogen and progesterone suppress the production of milk
- After birth, the pituitary begins to secrete *prolactin*, which then stimulates milk *production*
- The actual **release of milk from the breasts (lactation)** is mediated by **oxytocin**, which causes contractions within the mammary lobules; suckling motions send signals through nerve endings to the hypothalamus, which then stimulates the pituitary to release oxytocin.
- For some women, the sound of an infant crying can also stimulate the production of oxytocin.

Lactation



Fetal Development Review



Sexually Transmitted Infections

*you do not need to memorize individual STIs and their effects. You should be able to list some examples, and understand their implications for fertility and general health of both parents and fetus/baby

- Caused by viruses, bacteria, or parasites
- Transmitted through sexual contact
- Impact general health as well as the ability to reproduce

HIV/AIDS

Hepatitis B

Genital Herpes

Syphilis

Chlamydia

Gonorrhea

Human Papillomavirus*

Pelvic Inflammatory Disease



Teratogens

You should be able to describe some examples teratogens and their impact on embryonic or fetal development.

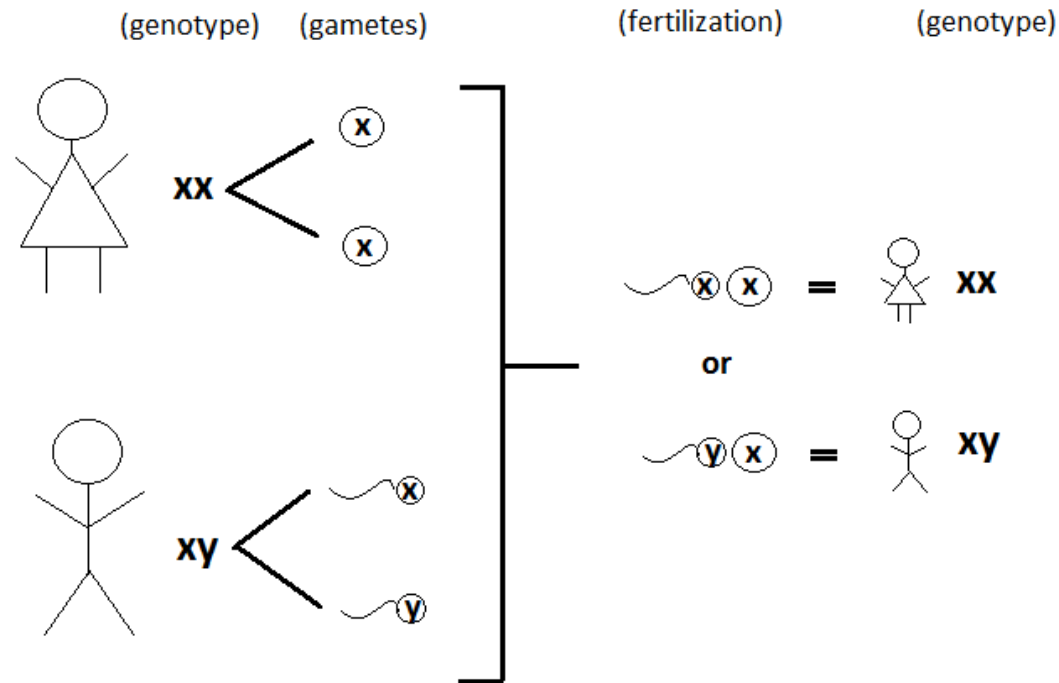
The placenta does NOT filter out harmful substances...

- Drugs, alcohol, etc. circulating in the mothers blood will pass through the placenta to the fetus's blood
- Can have severe consequences (such as FASD, or Fetal Alcohol Spectrum Disorder)
- Such substances are referred to as *teratogens*
- Teratogens are most harmful during the first trimester during when the neural tube and most organs develop
- Why would the developing blastocyst not be very sensitive to teratogens during the first 10-14 days after fertilization?
 - The cells in the blastocyst have not begun to differentiate yet so if one cell is damaged, a different one can easily take over

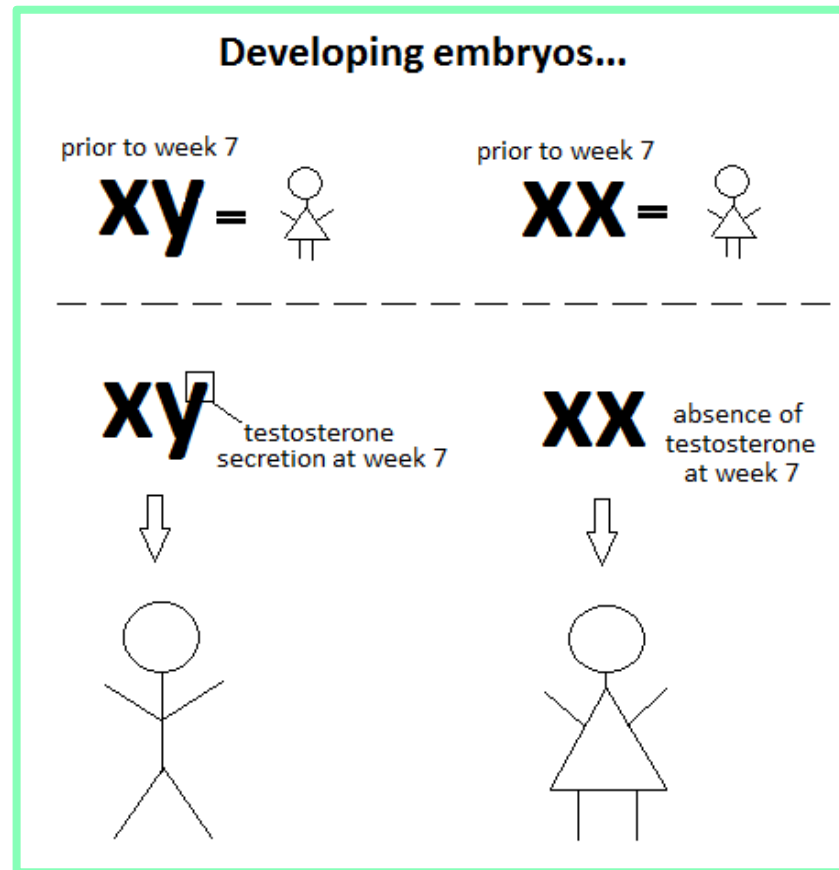
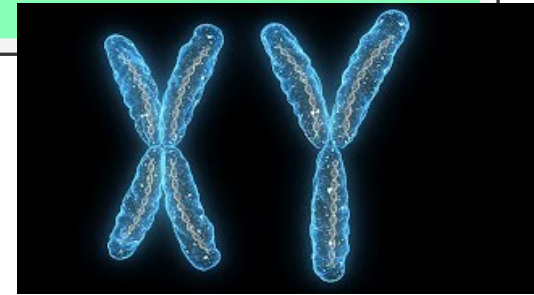
Sex Determination in Humans

➤ *Chromosomal (or genetic) sex* is determined at fertilization

- XX → Female
- XY → Male



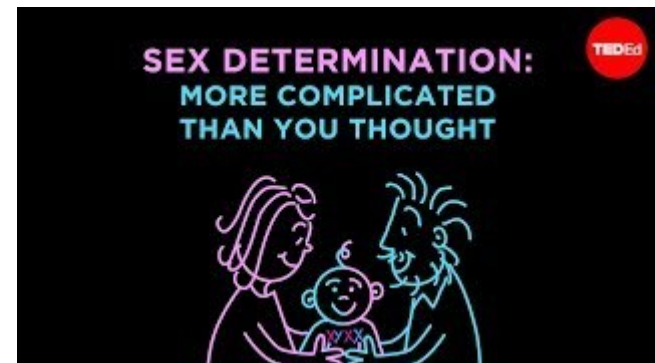
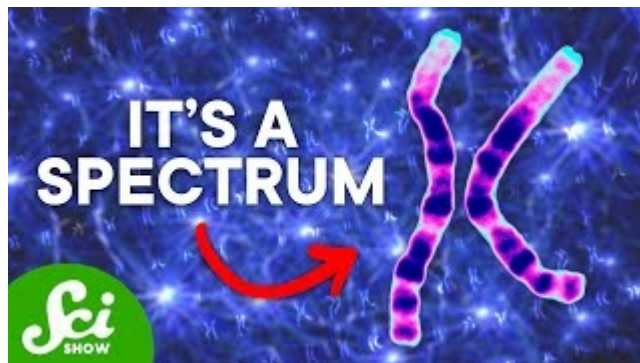
However, *anatomic sex* is not determined until about the seventh week of development:



- At week seven, genes contained within the short arm of the **Y chromosome encode the secretion of testosterone**, which then causes the embryo to begin differentiating.
- The absence of this gene means that no testosterone is produced. As a result, the embryo will continue developing as a female.



*Is it possible for a person to be genetically male
but anatomically female? Vice versa?*

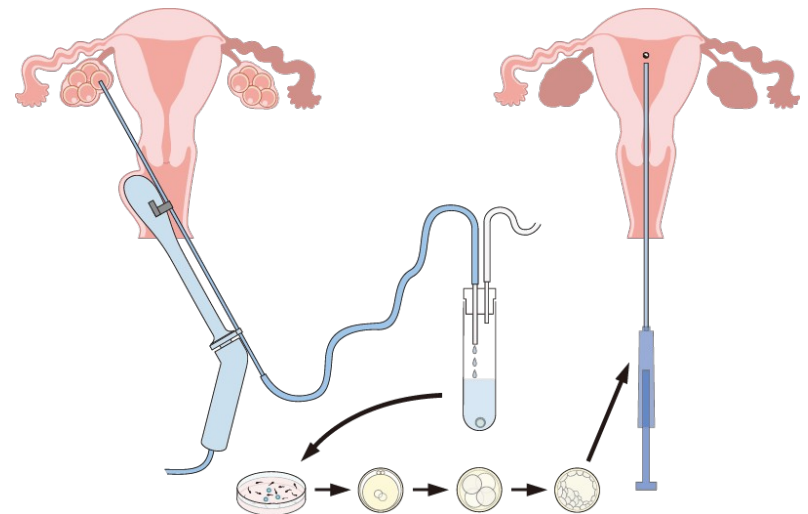
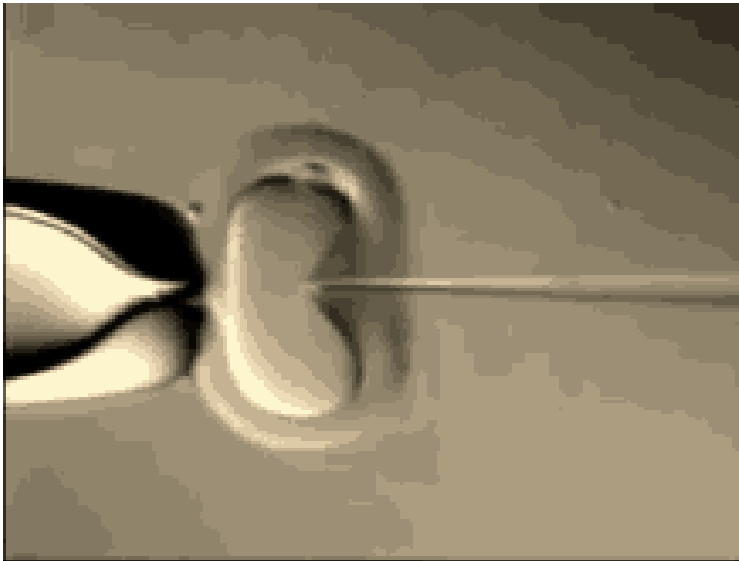


Reproductive Technologies

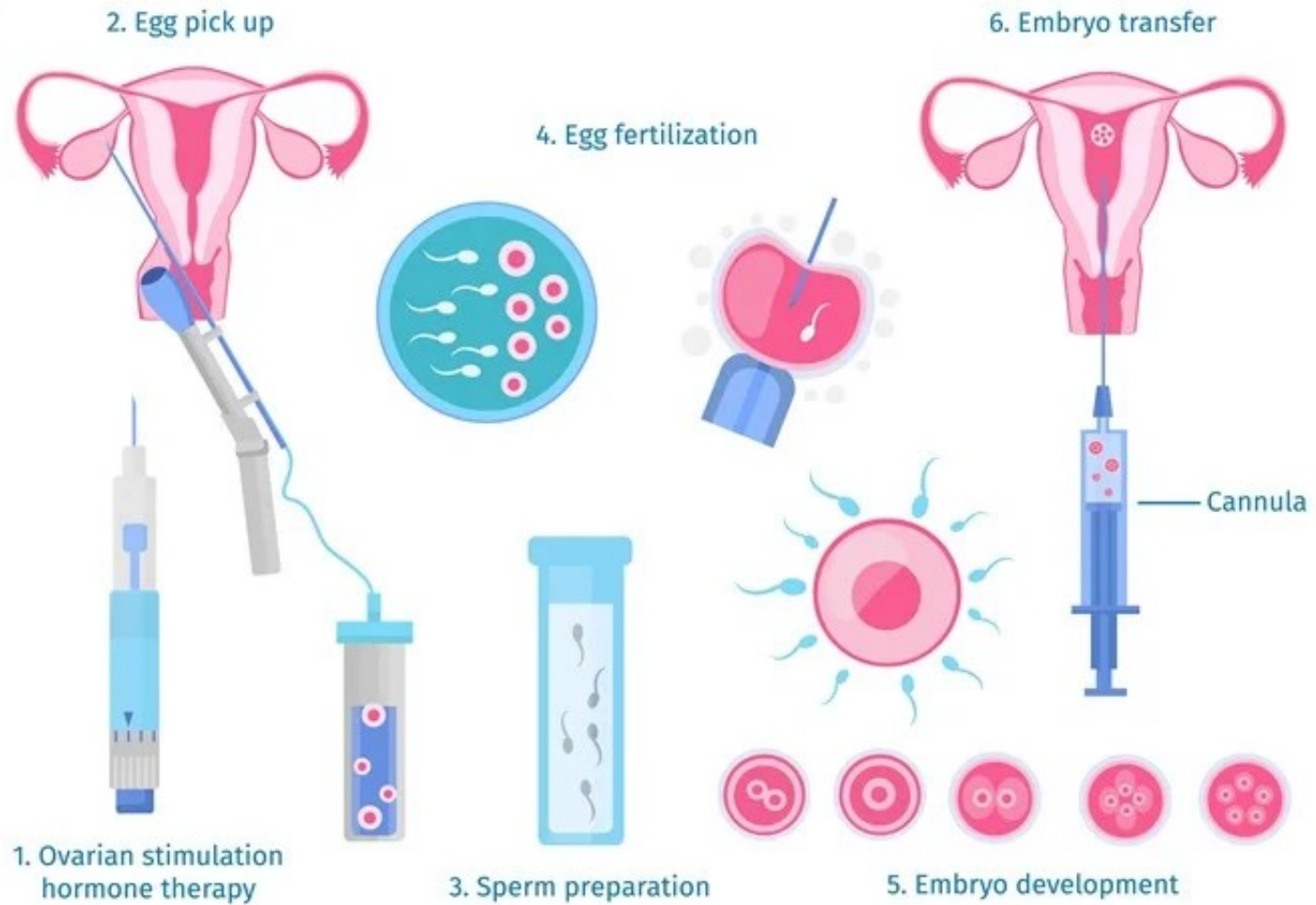
Technologies to support conception

In-vitro fertilization (IVF)

- involves retrieving eggs from a woman's ovaries and fertilizing them with sperm in a laboratory setting. The resulting embryos are then transferred directly into the woman's uterus, aiming to achieve pregnancy.



in Vitro Fertilization



Reproductive Technologies

Technologies to support conception

Fertility Drugs

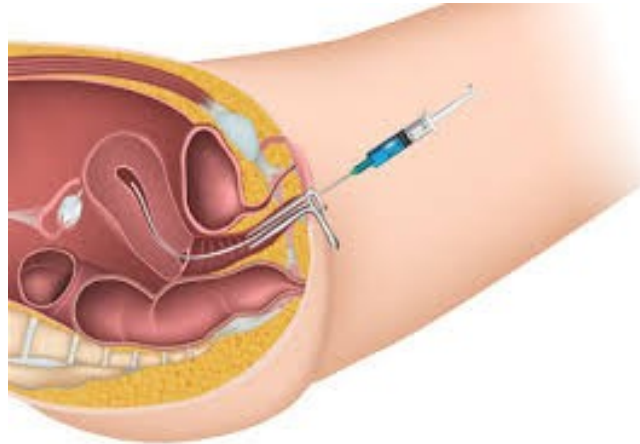
- medications used to stimulate ovulation and enhance the chances of conception in individuals facing fertility issues. They work by regulating hormones to promote the development of multiple eggs, increasing the likelihood of successful fertilization and pregnancy
- Typically a prerequisite step for successful IVF
- E.g. Clomiphene citrate. Taken by mouth, this drug stimulates ovulation by causing the pituitary gland to release more follicle-stimulating hormone (FSH) and luteinizing hormone (LH), which stimulate the growth of an ovarian follicle containing an egg.

Reproductive Technologies

Technologies to support conception

Artificial Insemination

- fertility treatment that involves directly inserting sperm into a woman's reproductive tract to facilitate conception without sexual intercourse.



Reproductive Technologies

Technologies to prevent conception

Physical barriers (i.e. male condoms and female diaphragms)

- Prevent sperm from reaching eggs
- Male condoms offer additional protection against Sexually Transmitted Infections (STIs)

Chemical barriers (e.g. spermicide)

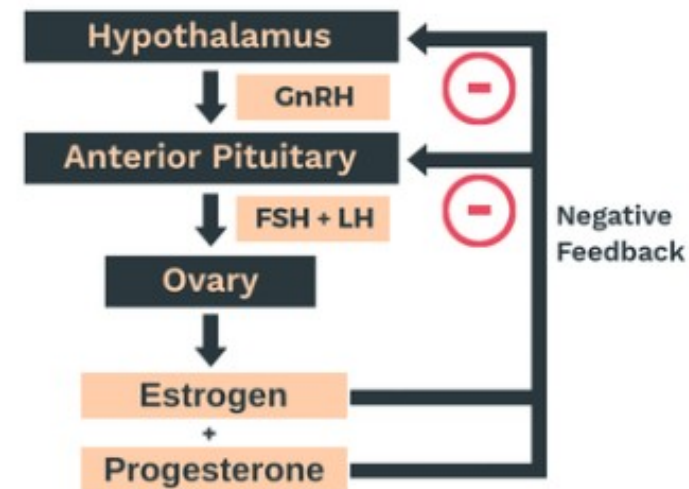
- Kill sperm and therefore prevent them from reaching and fertilizing an egg
- Lower effectiveness, typically used in combination with other methods

Negative Feedback and Hormonal Contraceptives



Hormonal Contraceptives (e.g. Birth control pill, injections, Intrauterine Devices (IUDs), etc.)

- Contraceptive technologies whereby artificial hormones mimic the effect of progesterone and therefore through **negative feedback** inhibit the release of FSH and LH from the anterior pituitary. This ultimately prevents ovulation.



Morning After Pill (i.e. Plan B)

- More intensive hormone treatment taken for emergency use
- Very high dose of synthetic hormones that can prevent or delay ovulation if taken within 48 hours of intercourse

Reproductive Technologies

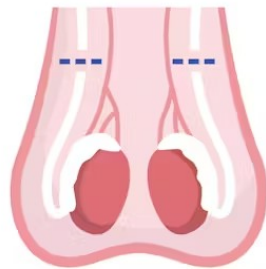
Technologies to prevent conception

Tubal Ligation

- Surgery that involves cutting the oviducts and tying off the cut ends. Prevents ovum from reaching sperm and the uterus.

Vasectomy

- Male equivalent to tubal ligation, whereby the vas deferens is cut and tied off. Sperm do not enter semen and leave the male body.



MALE STERILIZATION
VASECTOMY



FEMALE STERILIZATION
TUBAL LIGATION